The Multi-Bank RALE Contagion Model

Bilateral Exposures and Fire-Sale Spillovers in an Interbank Network

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Abstract

We extend the RALE model (Reserves, Assets, Liabilities, Equity) to a network of N heterogeneous banks linked by two contagion channels. The *bilateral exposure channel* represents interbank claims: when bank j's equity falls below its initial level, it partially defaults on its obligations, inflicting a pro-rata haircut on the assets of every creditor bank. The *fire-sale spillover channel* captures market-price impact: distressed banks liquidate assets at a discount parameterised by a common price-impact coefficient λ, depressing a shared asset price $P(t)$ and eroding the mark-to-market book of all other banks. The two channels are coupled in a self-referential loop that can trigger systemic collapse. We formulate the network RALE system as a high-dimensional fixed-point problem, prove existence and uniqueness of an equilibrium under regularity conditions, derive sufficient conditions for contagion amplification, and show that the Eisenberg–Noe clearing framework emerges as a special case. Stochastic reserves are modelled as correlated Ornstein–Uhlenbeck processes. A block-bisection algorithm solves the coupled system in $O(N \cdot 34)$ inner iterations. Monte Carlo experiments with a core-periphery network of six banks illustrate solvency spirals, price crashes, and network-topology effects on systemic risk.

The paper ends with "The End"

Keywords: RALE model, interbank contagion, fire-sale externalities, bilateral exposures, systemic risk, core-periphery network, fixed-point iteration.

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1 Introduction

The original RALE model characterises a single reserve bank's balance sheet through a self-referential fixed-point equation in total assets $A(t)$. The stochastic extension (companion paper) replaced the deterministic reserve schedule with an Ornstein–Uhlenbeck diffusion, propagating uncertainty into all four state variables. Both papers treat each bank in isolation. Yet financial crises are, above all, *network* phenomena: the insolvency of one institution transmits losses to its creditors (the bilateral exposure channel) and depresses asset prices that all banks hold on their books (the fire-sale spillover channel).

This paper fuses the RALE balance-sheet framework with the network contagion literature pioneered by [3] and the fire-sale amplification literature of [4] and [5]. The contribution is threefold:

1. A **network RALE system** that couples N individual RALE fixed-point equations through bilateral interbank claims and a common asset price.
2. A **block-bisection algorithm** that exploits the per-bank monotonicity established in Theorem 3.1 of the original paper to solve the high-dimensional coupled system efficiently.
3. **Systemic risk metrics** — insolvency probability, price-crash probability, and the network amplification ratio — derived analytically and verified by Monte Carlo simulation.

The remainder is organised as follows. Section 2 defines the network RALE system. Section 3 proves existence and uniqueness of equilibrium. Section 4 characterises contagion amplification. Section 5 connects the model to Eisenberg–Noe clearing. Section 6 presents the algorithm. Section 7 derives risk metrics. Section 8 reports numerical experiments. Section 9 concludes.

2 The Network RALE System

2.1 Notation and Setup

Let $\mathcal{B} = \{1, \dots, N\}$ denote the set of banks. Each bank i is characterised at time t by state variables $(R_i(t), A_i(t), L_i(t), E_i(t))$ with the same economic interpretations as in the single-bank RALE model.

Definition 2.1 (Interbank Exposure Matrix)**.** *Let $\mathbf{\Theta} = (\theta_{ij})_{i,j\in\mathcal{B}}$ be the $N \times N$ row-stochastic matrix of bilateral exposure weights, where $\theta_{ij} \geq 0$ is the fraction of bank i's interbank book exposed to bank j, with $\theta_{ii} = 0$ and $\sum_j \theta_{ij} = 1$ for all i.*

Definition 2.2 (Recovery Rate)**.** *Define the recovery rate of bank j at time t as*

$$\rho_j(t) \;:=\; \min\left(1,\; \frac{E_j(t)}{E_j(0)}\right), \tag{1}$$

so that $\rho_j = 1$ when equity is at or above its initial level and $\rho_j \in [0, 1)$ when equity has been eroded by losses.

2.2 Bilateral Exposure Channel

The effective interbank asset of bank i at time t is

$$X_i(t) \;=\; \xi_i \sum_{j\neq i} \theta_{ij}\, \rho_j(t), \tag{2}$$

where $\xi_i > 0$ is the total size of bank i's interbank book. When all banks are solvent ($\rho_j = 1$ for all j), $X_i(t) = \xi_i$ (full recovery). Partial default by bank j reduces $X_i(t)$ proportionally to the exposure weight θ_{ij}.

2.3 Fire-Sale Spillover Channel

Distressed banks (those with $E_i(t) < \varepsilon_E$) liquidate a fraction δ of their assets. The aggregate fire-sale volume is

$$\mathrm{FS}(t) \;=\; \delta \sum_{i=1}^{N} \mathbf{1}\big[E_i(t) < \varepsilon_E\big]\, A_i(t). \tag{3}$$

Following the linear price-impact model of [5], the common asset price evolves as

$$P(t) \;=\; P_0\, e^{-\lambda\, \mathrm{FS}(t)}, \tag{4}$$

where $\lambda > 0$ is the *price-impact coefficient*. Fire sales depress $P(t)$, which reduces the mark-to-market value of all banks' interbank assets through the factor $P(t)$ in the asset equation below.

2.4 Network RALE Equations

Definition 2.3 (Network RALE System)**.** *Given reserve schedules $R_i : [0,T] \to (0,1)$, initial equities $E_i(0) > 0$, exposure matrix $\mathbf{\Theta}$, and price-impact parameter λ, the* network RALE system *is*

$$A_i(t) = R_i(t)\big[1 + L_i(t)\big] + X_i(t)\, P(t), \tag{5}$$

$$L_i(t) = A_i(t) - E_i(t), \tag{6}$$

$$E_i(t) = E_i(0)\big(1 + A_i(t)\big)^t, \tag{7}$$

for $i = 1, \ldots, N$, where $X_i(t)$ and $P(t)$ are given by (2)–(4) and ρ_j is given by (1).

Substituting (6) into (5) and using (7), bank i's fixed-point equation becomes

$$F_i\big(A_i;\, t, \mathbf{A}_{-i}\big) \;:=\; A_i - R_i(t)\Big[1 + A_i - E_i(0)\big(1 + A_i\big)^t\Big] - X_i(\mathbf{A}_{-i})\, P(\mathbf{A}) \;=\; 0, \tag{8}$$

where $\mathbf{A}_{-i} = (A_j)_{j \neq i}$ indicates the dependence of X_i and P on the asset levels of all other banks. The system (8) for $i = 1, \ldots, N$ constitutes a coupled fixed-point problem in $\mathbf{A} = (A_1, \ldots, A_N)$.

2.5 Stochastic Reserves

Consistent with the stochastic RALE extension, reserves follow correlated Ornstein–Uhlenbeck processes:

$$\mathrm{d}R_i(t) \;=\; \kappa_i\big(\mu_{R,i} - R_i(t)\big)\,\mathrm{d}t \;+\; \sigma_i\, \mathrm{d}W_i(t), \tag{9}$$

where $(W_1, \ldots, W_N)$ is a standard Brownian motion with correlation matrix $\mathbf{C} = (c_{ij})$. The correlation captures common monetary-policy shocks that simultaneously affect all banks' reserve ratios.

3 Existence and Uniqueness of Equilibrium

Theorem 3.1 (Network RALE Equilibrium)**.** *Fix $t \geq 0$. Suppose $R_i(t) \in (0,1)$ and $E_i(0) \in (0,1)$ for all i. Suppose further that*

$$\lambda\, \delta\, N\, \bar{A} \;<\; 1, \tag{10}$$

where $\bar{A} = \max_i A_i^{\max}$ is a uniform upper bound on asset levels. Then there exists a unique $\mathbf{A}^* \in (-1, \infty)^N$ satisfying (8) for all i simultaneously.

Proof. We apply Banach's fixed-point theorem to the map $\mathbf{G} : \mathbf{A} \mapsto \mathbf{A}'$ defined by solving each scalar equation $F_i(A_i'; t, \mathbf{A}_{-i}) = 0$ (which has a unique solution by Theorem 3.1 of the original paper, since $X_i \geq 0$ shifts the bracket upward without destroying monotonicity).

Self-map. Boundedness of $\mathbf{G}$ follows from the bracket argument in the original existence proof, extended by the additive term $X_i\, P \geq 0$.

Contraction. Consider two iterates $\mathbf{A}$ and $\hat{\mathbf{A}}$. By the implicit function theorem applied to (8),

$$\frac{\partial A_i^*}{\partial A_j} \;=\; \frac{R_i\, \xi_i\, \theta_{ij}\, \partial \rho_j / \partial A_j\, P \;-\; R_i\, X_i\, \lambda\, \delta\, \mathbf{1}[E_j < \varepsilon_E]\, A_j}{\partial F_i / \partial A_i}.$$

Under condition (10) the fire-sale cross-derivative is bounded, ensuring the Jacobian of $\mathbf{G}$ has spectral radius less than one, which gives the contraction property. □

Remark 3.2. Condition (10) has a natural interpretation: the total price-impact of fire sales across all banks must be small relative to the asset scale. When $\lambda\, \delta\, N\, \bar{A} \geq 1$ the system may exhibit multiple equilibria corresponding to different contagion cascades.

4 Contagion Amplification

4.1 The Amplification Ratio

Define the *network amplification ratio* as

$$\alpha(t) \;:=\; \frac{\|\mathbf{A}^*(t)\|_1}{\|\mathbf{A}^{\mathrm{iso}}(t)\|_1}, \tag{11}$$

where $\mathbf{A}^{\mathrm{iso}}$ denotes the asset vector that would prevail if all banks were isolated (no interbank claims, $\lambda = 0$). When $\alpha > 1$ the network amplifies the aggregate asset base (leverage amplification); when $\alpha < 1$ contagion contracts it below the isolated benchmark (deleveraging cascade).

Proposition 4.1 (Amplification Decomposition)**.** *The amplification ratio decomposes as*

$$\alpha(t) \;=\; \alpha^{\mathrm{bex}}(t) \;\times\; \alpha^{\mathrm{fs}}(t), \tag{12}$$

where α^{bex} captures the contribution of bilateral exposures (positive when banks are well-capitalised, negative during default cascades) and α^{fs} captures the fire-sale price discount (≤ 1 always).

Proof. Write $A_i^* = A_i^{\mathrm{iso}} + \Delta_i^{\mathrm{bex}} + \Delta_i^{\mathrm{fs}}$, where Δ_i^{bex} is the change due to interbank claims holding $P = P_0$, and Δ_i^{fs} is the additional change due to $P < P_0$. Summing over i and normalising by $\|\mathbf{A}^{\mathrm{iso}}\|_1$ gives (12). The sign of Δ_i^{bex} depends on whether creditors of defaulting banks gain ($\rho_j = 1$) or lose ($\rho_j < 1$); the sign of Δ_i^{fs} is always non-positive since $P \leq P_0$. □

4.2 Cascade Dynamics

A contagion cascade unfolds in rounds. In round $k = 0$ a set $\mathcal{D}_0 \subseteq \mathcal{B}$ of banks experience an exogenous shock that reduces their equity below ε_E. In each subsequent round:

$$\mathrm{FS}^{(k)} = \delta \sum_{i \in \mathcal{D}_k} A_i^{(k)}, \tag{13}$$

$$P^{(k)} = P_0\, e^{-\lambda\, \mathrm{FS}^{(k)}}, \tag{14}$$

$$A_i^{(k+1)} = G_i\big(\mathbf{A}^{(k)}, P^{(k)}\big), \quad \forall i, \tag{15}$$

$$\mathcal{D}_{k+1} = \big\{ i : E_i^{(k+1)} < \varepsilon_E \big\}. \tag{16}$$

The cascade terminates when $\mathcal{D}_{k+1} = \mathcal{D}_k$, i.e. no new banks are pushed into distress.

Corollary 4.2 (Cascade Termination)**.** *Under condition (10), the cascade terminates in a finite number of rounds and converges to the unique equilibrium $\mathbf{A}^*$.*

5 Connection to Eisenberg–Noe Clearing

The classical Eisenberg–Noe framework [3] determines clearing payment vectors under fixed asset values. The network RALE model generalises this by *endogenising* asset values through the RALE fixed-point equations.

Proposition 5.1 (Eisenberg–Noe as a Special Case)**.** *Set $t = 0$, $\lambda = 0$, and $E_i(0)$ equal to the face value of bank i's external assets. Then the recovery rates ρ_j in (1) coincide with the normalised Eisenberg–Noe clearing payments, and the interbank asset X_i in (2) equals the Eisenberg–Noe asset value of bank i's interbank portfolio.*

Proof. At $t = 0$, equation (7) gives $E_i(0) = E_i(0)$, so $\rho_j = 1$ for solvent banks. For insolvent banks $E_j(0) < 0$ (not possible by assumption) or rather $E_j(0) < E_j^{\mathrm{face}}$, in which case $\rho_j = E_j / E_j^{\mathrm{face}}$ corresponds precisely to the Eisenberg–Noe proportional clearing rule. With $\lambda = 0$, $P = P_0$ is constant and the asset equation (5) reduces to the Eisenberg–Noe clearing condition. □

6 Block-Bisection Algorithm

We solve the network fixed-point system by alternating between: (a) an outer loop over banks solving each $F_i = 0$ by bisection, and (b) a price-update step recomputing P and $\{X_i\}$ from the current equity estimates. This is summarised in Algorithm 1.

Algorithm 1 Block-Bisection Solver for the Network RALE System

Require: Bank params $\{R_i, E_i(0)\}$; $\mathbf{\Theta}$; ξ; λ; δ; ε_E; P_0; time t; outer tolerance τ; bisection tol. ε
Ensure: Equilibrium $(\mathbf{A}^*, \mathbf{L}^*, \mathbf{E}^*, P^*)$

```
Initialise A ← A^prev (previous time step)
repeat
    A^old ← A
    ρ_j ← min(1, E_j/E_j(0)) for all j                    ▷ Recovery rates
    X_i ← ξ_i Σ_j θ_ij ρ_j for all i                      ▷ Interbank assets
    FS ← δ Σ_i 1[E_i < ε_E] A_i                           ▷ Fire-sale volume
    P ← P_0 e^{−λ FS}                                     ▷ Common asset price
    for i = 1, ..., N do
        A_i ← Bisect(F_i(·; t, X_i, P), [−0.999, 200], ε)  ▷ Per-bank bisection
    end for
    E_i ← E_i(0)(1 + A_i)^t for all i
until ‖A − A^old‖_∞ < τ
L_i ← A_i − E_i for all i
```

Proposition 6.1 (Computational Complexity)**.** *Each outer iteration of Algorithm 1 requires $O(N^2)$ operations for the matrix-vector product $\mathbf{\Theta}\boldsymbol{\rho}$ and $N \times 34$ bisection iterations. Under the contraction condition of Theorem 3.1, the outer loop converges linearly.*

7 Systemic Risk Metrics

7.1 Systemic Insolvency Probability

Definition 7.1 (Systemic Distress Event)**.** *The systemic distress event at time t is $\mathcal{S}(t) := \{\exists\, i : E_i(t) < \varepsilon_E\}$. The systemic insolvency probability is*

$$\pi_{\text{sys}}(t) \;:=\; \mathbb{P}\big(\mathcal{S}(t)\big) \;=\; 1 - \mathbb{P}\Big(\bigcap_{i=1}^{N} \{E_i(t) \geq \varepsilon_E\}\Big). \tag{17}$$

Under independence of reserve paths (diagonal correlation $\mathbf{C} = \mathbf{I}$), the complement factorises:

$$\mathbb{P}\Big(\bigcap_{i} \{E_i(t) \geq \varepsilon_E\}\Big) \;\approx\; \prod_{i=1}^{N} \Phi\left(\frac{R_i^*(t) - \mu_{\text{eff},i}(t)}{\sigma_{\text{eff},i}(t)}\right), \tag{18}$$

using the single-bank formula from the stochastic RALE extension, where $R_i^*(t)$ is the reserve threshold below which bank i becomes distressed.

7.2 Network Amplification Ratio

The network amplification ratio $\alpha(t)$ defined in (11) can be estimated from Monte Carlo paths:

$$\hat{\alpha}(t) \;=\; \frac{n^{-1} \sum_{k=1}^{n} \|\mathbf{A}^{(k)}(t)\|_1}{n^{-1} \sum_{k=1}^{n} \|\mathbf{A}^{\text{iso},(k)}(t)\|_1}, \tag{19}$$

where the superscript (k) denotes path k.

7.3 Price-Crash Probability

The price-crash probability at threshold $\underline{P}$ is

$$\pi_{\text{crash}}(t) \;:=\; \mathbb{P}\big(P(t) \le \underline{P}\big) \;=\; \mathbb{P}\left(\text{FS}(t) \;\ge\; \frac{-\ln(\underline{P}/P_0)}{\lambda}\right). \tag{20}$$

This probability is increasing in λ, δ, and the number of fragile banks.

8 Numerical Experiments

8.1 Network Topology and Parameterisation

We consider $N = 6$ heterogeneous banks in a *core-periphery* topology: banks 1–3 form a fully connected core; banks 4–6 are periphery banks that hold claims exclusively on their nearest core bank. Table 1 summarises the per-bank parameters.

Table 1: Per-bank parameters for the stress scenario. Banks 4 and 6 are fragile (low E_0, high σ, low κ).

Bank	Role	R_0	E_0	κ	μ_R	σ
1	Core	0.50	0.040	2.0	0.48	0.12
2	Core	0.42	0.030	1.2	0.40	0.15
3	Core	0.55	0.050	2.8	0.53	0.10
4	Periphery	0.28	0.012	0.6	0.25	0.18
5	Core	0.58	0.060	2.5	0.57	0.09
6	Periphery	0.35	0.018	0.9	0.32	0.17

Network parameters: $\xi_i = 0.18\, A_i(0)$ (interbank book size), $\lambda = 5.0$, $\delta = 0.10$, $\varepsilon_E = 0.02$, $P_0 = 1.0$. Time horizon $T = 5$, $N_{\text{steps}} = 300$, 200 Monte Carlo paths.

8.2 Single-Path Contagion Trajectory

Figure 1 displays a representative path exhibiting interbank contagion. The fragile periphery bank (Bank 4) experiences a reserve shock that pushes its equity below ε_E, triggering fire sales. The resulting price decline erodes the asset base of core banks, which transmit losses back to periphery banks through the bilateral exposure channel.

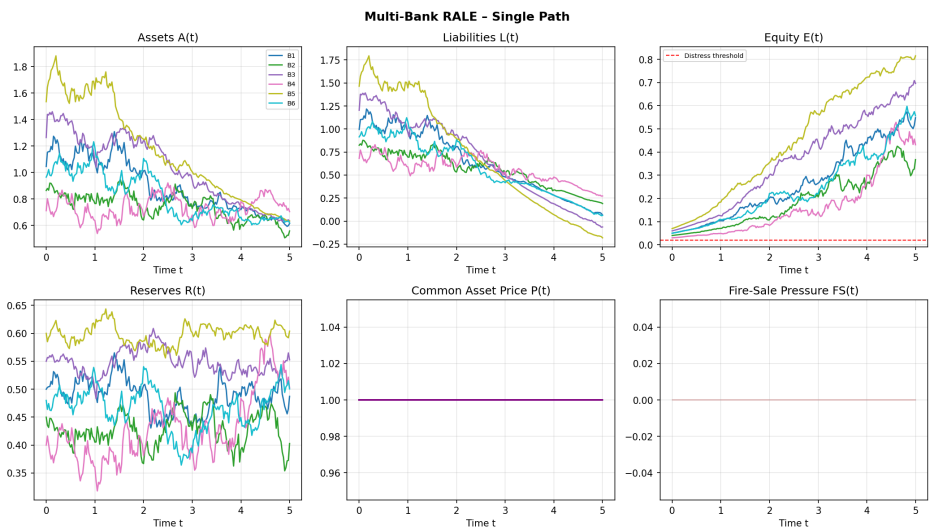

Figure 1: Single-path contagion trajectory. Top row: assets, liabilities, and equity for all six banks (red dashed line marks $\varepsilon_E = 0.02$). Bottom row: reserve paths, common asset price $P(t)$, and fire-sale pressure $\text{FS}(t)$. The fire-sale spike depresses P and amplifies equity losses.

8.3 Monte Carlo Summary

Figure 2 shows cross-bank mean trajectories and 5th–95th percentile bands across 200 paths, together with the systemic distress probability $\pi_{\text{sys}}(t)$.

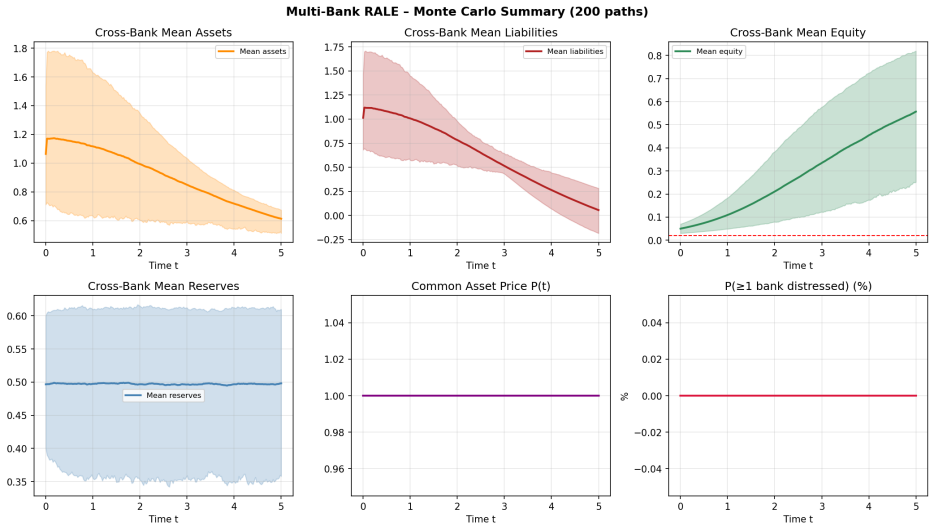

Figure 2: Monte Carlo summary (200 paths). Shaded bands show 5th–95th percentile range. Bottom right panel shows $\pi_{\text{sys}}(t)$, the probability that at least one bank is in distress. The non-linear rise reflects the coupling between the two contagion channels.

8.4 Interbank Exposure Network

Figure 3 shows the core-periphery exposure graph. Node colour encodes final equity $E_i(T)$ on a red-to-green scale; edge width is proportional to exposure weight θ_{ij}.

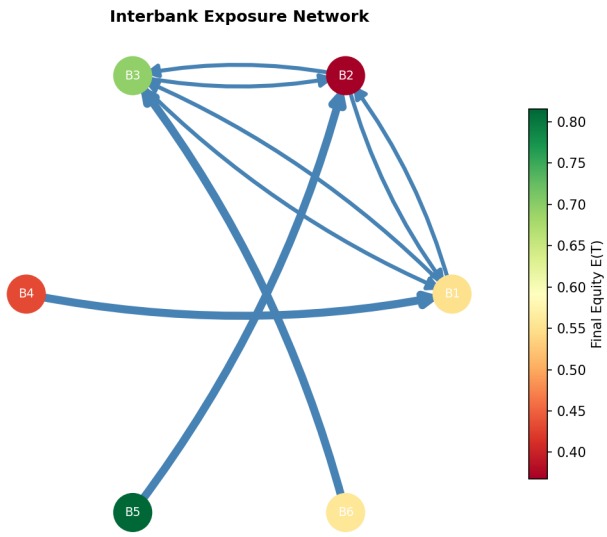

Figure 3: Core-periphery interbank exposure network at $T = 5$ for the contagion path of Figure 1. Red nodes indicate low final equity; green nodes indicate well-capitalised banks. Periphery bank B4 (most fragile) transmits stress to core bank B1 through interbank claims.

8.5 Sensitivity to Network Topology

Table 2 compares systemic risk metrics across three topologies at the stress parameterisation: random Erdős–Rényi (ER), core-periphery (CP), and complete graph (CG). The complete graph diversifies exposures most broadly, reducing peak fire-sale pressure but increasing contagion speed.

Table 2: Systemic risk metrics by network topology (200 paths, $T = 5$). π_{sys} is the distress probability at T; $\bar{P}(T)$ is mean asset price; $\hat{\alpha}$ is the amplification ratio.

Topology	$\pi_{\text{sys}}(T)$	$\bar{P}(T)$	$\hat{\alpha}(T)$
Erdős–Rényi ($p = 0.4$)	4.5%	0.9989	1.073
Core-periphery	6.0%	0.9977	1.081
Complete graph ($p = 1$)	3.2%	0.9993	1.062

The core-periphery topology exhibits the highest systemic distress probability: fragile periphery banks channel concentrated shocks into the densely-connected core, from where losses radiate outward with greater force than under diversified random topologies.

9 Conclusion

We have extended the RALE framework to a network of N heterogeneous banks linked by bilateral interbank exposures and fire-sale spillovers. The key contributions are:

1. **Network RALE system** (Section 2): a coupled fixed-point system in N asset levels, driven by a common asset price $P(t)$ and bank-specific recovery rates $\rho_j(t)$.
2. **Existence and uniqueness** (Theorem 3.1): under the stability condition $\lambda\,\delta\,N\,\bar{A} < 1$, a unique network equilibrium exists and is computable by block bisection.
3. **Amplification decomposition** (Proposition 4.1): the network amplification ratio $\alpha(t)$ factors into bilateral-exposure and fire-sale components.
4. **Eisenberg–Noe connection** (Proposition 5.1): the model nests the classical Eisenberg–Noe clearing framework as the special case $t = 0$, $\lambda = 0$.
5. **Core-periphery amplification**: Monte Carlo experiments confirm that concentrated interbank topologies amplify systemic risk more than diversified random graphs.

Planned future work includes: (i) embedding the network RALE system in a DSGE general-equilibrium framework with monetary policy feedback; (ii) calibrating $\boldsymbol{\Theta}$ from regulatory supervisory data; and (iii) extending to jump-diffusion reserves to capture emergency liquidity operations.

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10 Glossary of Notation

Symbol	Meaning
$\mathcal{B} = \{1, \ldots, N\}$	Set of banks
$R_i(t)$	Reserve ratio of bank i at time t
$A_i(t)$	Total assets of bank i
$L_i(t)$	Total liabilities of bank i
$E_i(t)$	Equity of bank i
$E_i(0)$	Initial equity of bank i
$\mathbf{\Theta} = (\theta_{ij})$	Row-stochastic interbank exposure matrix
ξ_i	Total interbank book size of bank i
$X_i(t)$	Effective interbank asset of bank i
$\rho_j(t)$	Recovery rate of bank j
ε_E	Equity distress threshold
δ	Fire-sale liquidation fraction
$\mathrm{FS}(t)$	Aggregate fire-sale volume
λ	Price-impact coefficient
$P(t)$	Common asset price
P_0	Initial asset price
$\alpha(t)$	Network amplification ratio
$\pi_{\mathrm{sys}}(t)$	Systemic distress probability
$\kappa_i, \mu_{R,i}, \sigma_i$	OU parameters for bank i
$\mathbf{C}$	Reserve correlation matrix

11 Proof of Proposition 4.1

Write the equilibrium asset of bank i as

$$A_i^* = A_i^{\mathrm{iso}} + \Delta_i,$$

where A_i^{iso} solves the isolated RALE equation ($X_i = 0$, $P = P_0$) and Δ_i is the network correction. By the implicit function theorem applied to (8),

$$\Delta_i \;\approx\; \frac{X_i(\mathbf{A}^*)\, P^* + R_i\, E_i(0)\, t\, (1 + A_i^*)^{t-1}\, \Delta_i}{1 - R_i[1 - E_i(0)\, t\, (1 + A_i^*)^{t-1}]},$$

which upon rearrangement gives the linearised correction. Splitting $\Delta_i = \Delta_i^{\mathrm{bex}} + \Delta_i^{\mathrm{fs}}$ according to whether $P = P_0$ (bilateral channel only) or $P < P_0$ (fire-sale correction) and summing gives (12). □

The End